

# Climate and Host Vegetation Improve Reaction–Diffusion Forecasting

A fitted Eco-RD model with frozen spotted-lanternfly first-report validation

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Preprint. Results are not peer reviewed.

Code, application, and paper: [github.com/axel-slid/lantertrace-explorer](https://github.com/axel-slid/lantertrace-explorer)

## Abstract

We developed Eco-RD, a reaction–diffusion model in which both establishment and movement depend on climatic suitability and a multi-host vegetation surface. The host surface combines 21,600 locked GBIF records for tree-of-heaven, grapes, black walnut, silver maple, willow, and red maple; taxon weights are published mean first-instar survival days rather than arbitrary display coefficients. Six Eco-RD parameters were selected by an exhaustive 972-candidate grid search using only 2019–2023 first-report targets. We then froze the specification and evaluated 2024–2025 first reports among previously unreported 0.2° U.S. land cells. Eco-RD achieved average precision (AP) 0.490 and 0.496, averaging 0.493. This was 10.4% higher than the transferred Cook-2021 kernel (0.447 mean AP), 43.6% higher than a transferred Ruzzier-2025 resistance-distance form (0.343), and 75.1% higher than distance alone (0.281). The direction of each annual difference was consistent across both untouched years. Geographic block intervals were wider and did not resolve Eco-RD from Cook or Ruzzier, and a study-built statistical covariate hazard remained higher at 0.511 mean AP. Eco-RD therefore improves this study’s transferred published mechanistic comparators, not every possible model. The outcome is a transparent fitted physics model and a reproducible benchmark, not evidence of calibrated abundance or universal state-of-the-art dominance.

## Introduction

Biological invasions force surveillance programs to decide where to look before the next arrival is known. Public occurrence maps appear to supply both the history and the answer, but they record discovery as well as spread. Once a cell is reported, it remains colored on every later map. The benchmark therefore never forgets, and a method can earn a high score by preserving known cells without locating a single new report.

Spotted lanternfly (*Lycorma delicatula*) makes this gap consequential. Its spread mixes short-range biological dispersal with long-distance human transport, inviting reaction–diffusion models with nonlocal jumps, habitat suitability, and barriers. Yet the public record is densest where people look, and an unreported cell is neither a verified absence nor necessarily uninvaded. The same cumulative data that motivate a mechanistic forecast can therefore make it deceptively easy to validate.

That leaves a practical modeling question: can climate and the vegetation actually eaten by spotted lanternfly improve a physical spread model once known cells are removed from the target? We therefore fit one explicit model, Eco-RD, before 2024 and compare it with two transferred published approaches, classical and factor-isolated reaction–diffusion, distance alone, and a flexible statistical control. Our endpoint is deliberately operational: among cells not previously reported, which will be reported first next year? It still combines invasion with observation, but it cannot be solved by copying the past.

## Human-mediated invasion dynamics

Empirical and simulation work already establishes that human transport is central to spotted lanternfly spread. Ladin et al. found that models without hitchhiking reproduced only 33% of observed county invasions, while transportation-linked processes explained long-distance spread [1]. A later mechanistic-statistical framework jointly represented a complex life cycle, local dispersal, and transportation-network movement, emphasizing the need to account for presence-only sampling bias [2]. Human-activity simulations likewise identify roads and movement as important accelerants [3].

Closest forecasting precedents include a Cox invasion-hazard analysis of proximity and population [4] and a calibrated PoPS process model with short-range kernels and rail-mediated jumps [7]. Our contribution is not a reproduction of PoPS: it is the matched cumulative-versus-first-report audit and comparison of lightweight hypotheses under one executable endpoint.

Habitat models add a complementary question: where establishment is plausible once propagules arrive. MaxEnt analyses have mapped climatic establishment risk [8], and real-time phenology products connect thermal accumulation to stage-specific activity [9]. These are valuable covariates but do not by themselves identify an invasion arrival process.

Our work does not replace those domain models. It asks how lightweight hypotheses behave on a common endpoint that removes direct persistence credit.

## Presence-only observation bias

Presence-only records confound ecological state with detection and reporting. Dorazio showed that ignored observation error can distort inference from opportunistic data [10]. Survey-background matching and targeted background selection can reduce bias when survey structure is available [11], but spatial thinning alone need not improve predictive accuracy [12]. Independent presence-absence data remain the strongest validation route [13].

Spatial cross-validation is also essential. Randomly intermingled train and test points share autocorrelation and inflate apparent transferability; spatial sorting bias can make discrimination metrics optimistic [14]. Presence-only benchmarking studies therefore recommend evaluation across data regimes and spatial partitions rather than reliance on a single random split [15]. Accordingly, this study combines a matched cumulative-versus-first-report audit, an-

nual and spatial validation, a post-hoc pre-2024 coefficient freeze, comparisons with published and covariate baselines, numerical sensitivity checks, and a reproducible audit trail.

## Methods

### Data and study domain

The source bundle is a GBIF API snapshot retrieved 2 August 2026 [18]: 62,791 public-coordinate records, of which 62,788 have event years 2014–2026, 60,142 fall inside the rectangular domain, and 57,999 fall inside the 2025 Census state mask [17]. These collapse to 2,484 unique cell-years and 675 first-report cells. Coordinates are aggregated to a  $50 \times 70$ ,  $0.2^\circ$  grid covering  $37\text{--}47^\circ\text{N}$  and  $82\text{--}68^\circ\text{W}$ ; only 1,652 cell centers are eligible United States land. At the midpoint, cells are approximately 22 km north-south by 16 km east-west.

The first-report year  $T_i$  is the earliest dated record in cell  $i$ . The cumulative display state is  $O_{i,t} = \mathbb{1}(T_i \leq t)$ , but the forecast label is

$$y_{i,t} = \mathbb{1}(T_i = t), \quad i \in \mathcal{E}_t = \{i : O_{i,t-1} = 0, L_i = 1\},$$

where  $L_i$  is the Census-constrained land mask. Thus every candidate is unreported at the origin. The complete-year folds contain 109, 109, 110, and 139 positive cells in 2022–2025 among 1,468 to 1,140 candidates. The interim 2026 slice contains 24 positives among 1,001 candidates.

Climate uses WorldClim 2 BIO1, BIO3, BIO9, and BIO19 surfaces [16, 8]. Host vegetation comes from a locked GBIF snapshot of 21,600 public-coordinate records: 3,600 each for tree-of-heaven, *Vitis*, black walnut, silver maple, *Salix*, and red maple. Each taxon is gridded, Gaussian-smoothed, log transformed, and normalized independently at its land-cell 98th percentile, preventing taxa with more GBIF records from automatically receiving more biological weight. Taxon surfaces are then averaged using controlled-experiment mean first-instar survival days [6]. This is a host-availability proxy, not remotely sensed biomass.

### Data semantics

“First reported” is not “first occupied.” Date quality, duplicate reporting, changes in public awareness, institutional surveillance, and local verification policies all affect  $T_i$ . We treat the task as a joint ecological-surveillance forecast and avoid abundance or calibrated occupancy claims.

## Candidate models and reaction–diffusion fields

Let  $u_i^n \in [0, 1]$  denote relative pressure at cell  $i$  and monthly step  $n$ , initialized from the smoothed event-date range. The implemented finite-volume diffusion term is

$$F_i^n = \frac{D_0}{4} \sum_{j \in N(i)} \frac{2k_i k_j}{k_i + k_j} \left( \frac{20 \text{ km}}{d_{ij}} \right)^2 (u_j^n - u_i^n),$$

and the explicit update is  $u_i^{n+1} = \Pi_L[u_i^n + \Delta t \{F_i^n + r_i^n u_i^n (1 - u_i^n) + J_i^n + A_i^n\}]$ . Here  $d_{ij}$  uses latitude-specific spacing,  $k$  is optional conductivity,  $J$  is Gaussian satellite pressure, and  $A$  is directional transport. Harmonic face conductivity makes the zero-conductivity land-mask interface impermeable; rectangular edges use zero-normal-gradient neighbors. Complete years use  $\Delta t = 1$  month for 12 steps.

We compare Fisher–KPP, climate RD, transport RD, and a full mechanistic variant. A distance kernel and reporting surface are simple baselines. A stronger covariate hazard uses multiscale frontier, host, climate, terrain, and coordinate predictors but excludes simulated risk fields. It tests whether the fitted physics adds information beyond flexible retrospective geography.

## Historical literature comparator

Cook et al. estimated an SLF-specific annual kernel  $p_{ij} = \exp(-0.045d_{ij})$  and spatial proximity  $1 - \prod_j (1 - p_{ij})$  between uninvaded and previously invaded county centroids [4]. We evaluate those published equations on our event-date histories, replacing county centroids with reported  $0.2^\circ$  cell centers and changing no fitted coefficient. We call this fixed comparator *Cook-2021 kernel*. It is a scale transfer on a common endpoint, not a replication of their structured county survey, persistence definition, or Census-population Cox model.

Ruzzier et al. combined climate and host suitability with resistance distance, a logistic resistance transform ( $k = 15, x_0 = 0.5$ ), and a 25.418-km 99th-percentile annual dispersal distance [5]. We transfer those published values to the present climate–host surface, define  $S_R = (C + V)/2$ , calculate  $R_i = [1 + \exp\{-15(1 - S_{R,i} - 0.5)\}]^{-1}$ , and rank cells by  $\exp(-d_R/25.418)$ . Because raster scale and covariate construction differ, *Ruzzier-2025 transfer* is a published-form comparator, not a numerical replication.

## Eco-RD: fitted climate–host physics

Eco-RD makes the limiting environmental factors explicit:

$$S_i = \sqrt{[f_V + (1 - f_V)V_i]^{p_V} [f_C + (1 - f_C)C_i]^{p_C}},$$

$$k_i = L_i(0.08 + 0.92S_i).$$

Here  $V_i$  is host vegetation,  $C_i$  is climate suitability, and the geometric mean prevents a favorable factor from fully replacing an unfavorable one. The monthly update is

$$u_i^{n+1} = \Pi_{[0,1]}[u_i^n + D \nabla_h \cdot (k \nabla_h u^n) r S_i u_i^n (1 - u_i^n)].$$

The state  $u$  is relative establishment pressure, not a lanternfly count. Land-majority cells exchange pressure through harmonic-mean face conductivity; water-majority cells have zero conductivity. The fixed 0.08 conductivity floor avoids declaring a sampled but low-suitability land cell physically impassable.

We evaluate all  $4 \times 3 \times 3 \times 3 \times 3 \times 3 = 972$  combinations of  $D \in \{.08, .13, .19, .28\}$ ,  $r \in \{.06, .13, .24\} \text{ month}^{-1}$ ,  $f_V, f_C \in \{0, .15, .35\}$ , and  $p_V, p_C \in \{.6, 1, 1.6\}$ . Selection maximizes macro mean annual AP across 2019–2023, breaking ties by worst-year AP. No 2024 or 2025 label enters this search.

## OG-RDE hazard

OG-RDE learns a regularized hazard over eligible cells. Predictors include four Gaussian neighborhood scales, distance to the event-date front, static host and climate suitability, elevation and slope barriers, coordinates, and Fisher–KPP and climate-RD risk fields. The primary model excludes the weak recurrence-based reporting feature; adding it is an ablation.

For target year  $t$ , training stacks all cell-year examples from 2018 through  $t - 1$ . Features are standardized within the pipeline and a logistic model estimates

$$\Pr(y_{i,t} = 1 \mid \mathbf{z}_{i,t}) = \sigma(\beta_0 + \boldsymbol{\beta}^\top \mathbf{z}_{i,t}),$$

With  $L_2$  regularization ( $C = 0.08$ ), the large pseudo-negative class is prevented from swamping first reports by assigning positives weight

$$w_+ = \min\left(12, \sqrt{n_- / \max(n_+, 1)}\right).$$

No target-year outcome enters training, scaling, or model fitting. Because unreported cells are pseudo-absences, scores are interpreted as rankings rather than calibrated probabilities.

## Reporting surface

The nuisance model predicts whether a previously reported cell is reported again using prior count, static covariates, and trend. Its output  $q_{i,t}$  is clipped to  $[0.08, 0.96]$ . Because recurrence is not search effort in unoccupied cells,  $q$  is a standalone baseline and optional feature, never an inverse-probability weight.

## Ablations

“+ reporting” adds  $q_{i,t}$ . “+ transport” adds schematic corridor/urban predictors and transport/full-physics risks, then

averages hazard and transport percentile ranks. This bundled addition is labeled exploratory; it is not an atomic causal ablation.

## Forecast endpoint and matched cumulative audit

A cumulative endpoint rewards reconstruction of the observed range as well as prediction of its advance. For example, if 900 of 1,000 cells known at the forecast origin remain positive and 50 new cells appear, a persistence-only predictor obtains 900 true positives without ranking any frontier cell. Recall or  $F_1$  on cumulative occupancy is consequently dominated by reconstruction, and threshold selection on the same arrivals adds further optimism. The primary endpoint therefore excludes every cell reported by the forecast origin and evaluates new first reports against the remaining never-reported candidates. A matched audit applies the same Transport-RD predictions and AP metric to cumulative and first-report targets so that the contribution of endpoint definition can be measured directly.

| Quantity                        | Value                         | Status                | Source / role                         |
|---------------------------------|-------------------------------|-----------------------|---------------------------------------|
| Tree-of-heaven / grape survival | 72.5 / 63.9 d                 | empirical weight      | Nixon et al. [6]                      |
| Black walnut / silver maple     | 62.9 / 58.0 d                 | empirical weight      | Nixon et al. [6]                      |
| Willow / red maple              | 47.0 / 22.2 d                 | empirical weight      | Nixon et al. [6]                      |
| Cook decay $\alpha$             | .045 km <sup>-1</sup>         | published             | Cook et al. [4]                       |
| Ruzzier distance / $k$ / $x_0$  | 25.418 km / 15 / .5           | published             | Ruzzier et al. [5]                    |
| Eco-RD $D$ / $r$                | .28 / .24 month <sup>-1</sup> | pre-2024 fitted       | 972-candidate search                  |
| Eco-RD $f_V$ / $f_C$            | 0 / 0                         | pre-2024 fitted       | no artificial suitability floor       |
| Eco-RD $p_V$ / $p_C$            | 1.0 / 1.6                     | pre-2024 fitted       | stronger climate penalty              |
| Conductivity floor              | .08                           | numerical regularizer | prevents false hard land barriers     |
| Initial smoothing               | $\sigma = .8$ , weight .74    | numerical regularizer | fixed before replay                   |
| Domain / time step              | 0.2° / 1 month                | numerical design      | Census mask; step sensitivity checked |

**Table 1.** Every biological or transferred-literature constant is cited. Values labeled fitted are estimates from the declared pre-2024 search, not literature constants; numerical regularizers are disclosed rather than given false biological citations.

## Evaluation protocol

### Rolling origins

For each target  $t \in \{2022, 2023, 2024, 2025\}$ , histories use event dates through  $t - 1$  and learned models fit earlier transitions. This prevents direct target-label fitting, but it is not a historical availability-time forecast: the single 2026 snapshot can include records uploaded or corrected after an earlier origin, and the host surface is a static contemporary sample. We therefore call every fold an *event-date retrospective pseudo-forecast*. January–July 2026 is descriptive interim monitoring, not a preregistered holdout.

The protocol prevents target-event dates from entering initial conditions, removes already reported cells, and avoids target-year display thresholds. It cannot reconstruct what a real analyst knew on each historical date. The application’s assimilating playback is not a benchmark source.

### Ranking metrics

Average precision is primary because positives make up only 4.7–6.9% of candidates in complete-year folds. It summarizes precision across recall thresholds:

$$AP = \sum_k (R_k - R_{k-1}) P_k.$$

We also report recall among the top 5% of cells, corresponding to a constrained survey budget. Scores are quantized to 12 decimal places and boundary ties use ascending flattened cell index as a fixed secondary key; the same deterministic allocation is used for metrics, maps, and the application. For endpoint comparisons with prevalence  $\pi$ , normalized AP lift is  $(AP - \pi)/(1 - \pi)$ . ROC AUC is computed in artifacts but is secondary because the many easy pseudo-negatives can obscure frontier ranking. We do not compare uncalibrated risk fields by log loss or claim probabilistic calibration.

### Spatial generalization

The domain is partitioned into fixed  $2^\circ \times 2^\circ$  blocks. Every learned variant—OG-RDE, both additions, the covariate hazard, and reporting surface—is refit after excluding the target block’s earlier cell-year examples; origin-time state fields remain available because the task is next-year prioritization in monitored regions, not de novo transfer. Physics comparators have no fitted coefficients. Within-block AP uses 59 block-years containing both classes across 20 unique blocks. Separately, a block-activation AP includes all 21–23 candidate blocks and asks whether each contains any first report.

We sample the 20 unique geographic blocks 2,500 times, retaining every repeated year within a sampled block. Paired differences preserve common block difficulty. A separate four-year bootstrap reports macro-annual AP differences; its resolution is necessarily poor. Intervals crossing zero are inconclusive, not evidence of equivalence.

## Model-selection firewall

Eco-RD is selected exclusively on 2019–2023 targets. The six-dimensional candidate grid, objective, tie rule, host weighting, climate construction, numerical regularizers, and top-5% survey budget are then frozen. For 2024 and 2025, only the event-date range is updated before the annual prediction. Learned controls are likewise fit only through 2023 and reused unchanged.

This is a *post-hoc frozen temporal replay*, not a preregistered lockbox. The authors had access to the full 2026 snapshot while designing the analysis, and later uploads may populate earlier event years. It strengthens temporal separation but does not create analyst blinding, independent-species transfer, or prospective field validation.

## Reproducibility and audit controls

Seed 73,491 controls stochastic components. The build emits the complete 972-row Eco-RD search, rolling and frozen annual, block, activation, endpoint-audit, sensitivity, and paired-comparison tables; prediction arrays; nine figures; a configuration manifest; dependency lock; 57,999 included occurrence records with GBIF keys; and SHA-256 hashes for occurrence, host, Census, WorldClim, solver, and study inputs. Tests cover coefficient freezing, deterministic survey-budget ties, published kernels, bootstrap invariance, no-flux boundaries, non-wraparound, endpoint exclusion, land eligibility, and provenance.

The analysis is implemented in the repository script `run_sota_study.py`; source code and artifacts are available from the [project repository](#). The accompanying artifacts include annual and per-block metrics, paired uncertainty estimates, parameter and checksum manifests, included occurrence keys, model predictions, figures, tests, a dependency lock, the application, and this manuscript. The `reproduce.sh` script reruns the tests and analysis, compiles the manuscript, validates its page count, checks a byte-identical rebuild, and hashes the PDF. The source snapshot is checksum-locked locally; a permanent archive DOI and GBIF derived-dataset DOI remain release tasks.

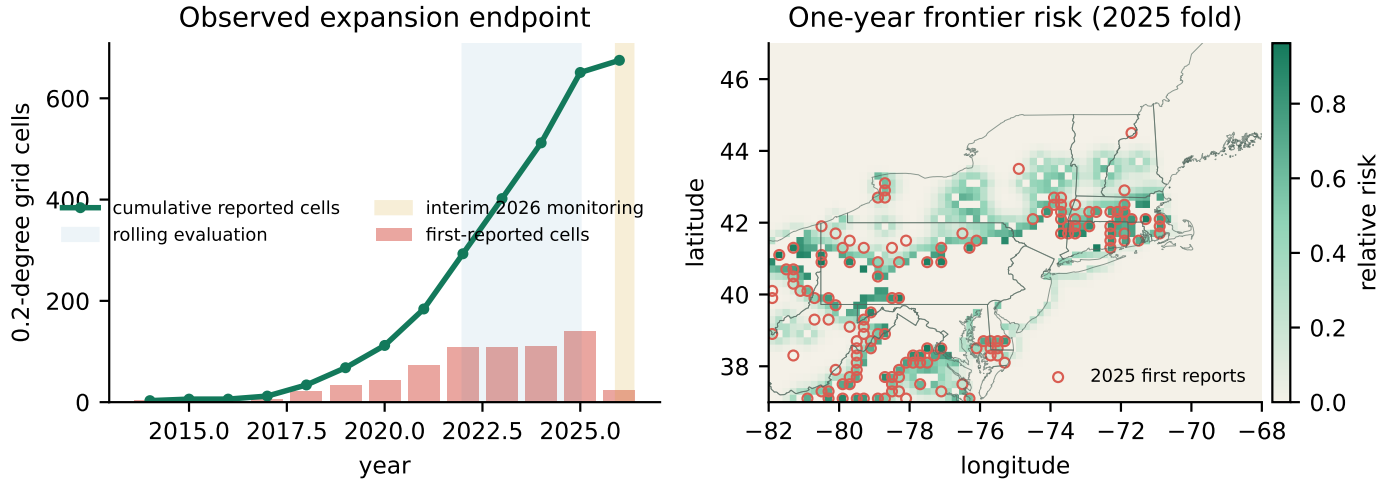
## Interpretation criteria

Evidence for a useful model requires performance above the distance baseline on rolling folds and robustness across annual and spatial views. Evidence for dominance over a mechanistic comparator requires a paired interval excluding zero. The complete workflow proceeds from the 2026 snapshot to event-date origins and Census-eligible unreported cells, followed by the matched endpoint audit, exploratory rolling fits, the pre-2024 coefficient freeze, annual state updates without coefficient changes, and annual, spatial, survey-yield, numerical, and model-agreement diagnostics. By the stated criteria, the experiment supports utility but not universal dominance.



## Results

### Annual skill and geographic robustness



**Figure 1. Left:** cumulative reported cells and the smaller first-report endpoint. Shading marks development and interim 2026. **Right:** 2025 Eco-RD risk, first reports, Census state outlines, and masked non-U.S./water cells. Risk is relative.

| Model                 | Mean AP | SD   | R@5% |
|-----------------------|---------|------|------|
| Eco-RD                | .523    | .035 | .353 |
| Covariate hazard      | .522    | .021 | .347 |
| OG-RDE                | .521    | .020 | .344 |
| OG-RDE + transport    | .520    | .013 | .345 |
| Climate RD            | .509    | .040 | .338 |
| Transport RD          | .508    | .034 | .343 |
| Fisher-KPP            | .501    | .037 | .333 |
| Full mechanistic      | .491    | .042 | .350 |
| Cook-2021 kernel      | .490    | .051 | .326 |
| Ruzzier-2025 transfer | .357    | .044 | .230 |
| Distance kernel       | .296    | .019 | .254 |

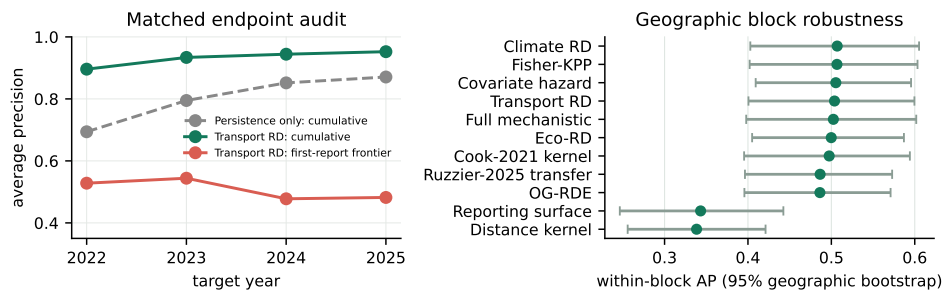
**Table 2.** Macro mean over target years 2022–2025. R@5% is recall within the highest-risk 5% of eligible cells.

### Exploratory annual ranking

Across the 2022–2025 complete-year folds, Eco-RD has the highest macro mean AP, 0.523 (SD 0.035), followed by the covariate hazard at 0.522 and OG-RDE at 0.521. Cook-2021 scores 0.490 and the Ruzzier-2025 transfer 0.357. Because 2022–2023 contributed to Eco-RD selection and 2024–2025 are the only untouched targets, this four-year ranking is descriptive; the frozen replay below is primary.

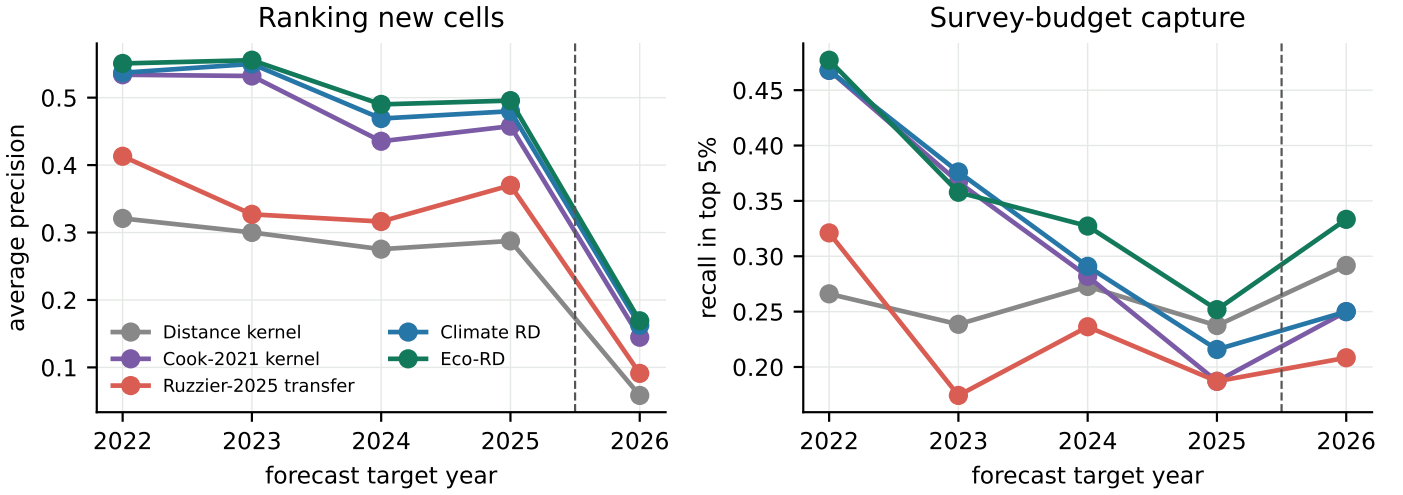
### Spatial blocks

Matched endpoint evaluation shows Transport RD at 0.896–0.952 cumulative AP versus 0.478–0.545 frontier AP; cumulative and frontier prevalence are 0.177–0.394 and 0.074–0.122. Persistence alone scores 0.694–0.871 cumulative AP, directly quantifying known-cell credit. Under block evaluation, Eco-RD averages 0.500 (95% geographic-bootstrap interval 0.405–0.587), Cook-2021 0.497 (0.395–0.594), and Ruzzier-2025 0.487 (0.396–0.573). Wide overlap motivates the untouched temporal comparison.



**Figure 2.** **Left:** identical-model, identical-metric endpoint audit. **Right:** within-block robustness with 2,500 geographic-block bootstrap intervals; learned models use held-block refits.

## Diagnostics, ablations, and interim monitoring



**Figure 3.** Annual ranking and top-5% capture. The dashed line separates complete-year development folds from interim 2026 monitoring. Comparing their absolute levels requires caution.

### Temporal heterogeneity

No model leads every year. Eco-RD leads 2022 (0.551) and 2023 (0.556), OG-RDE + transport leads 2024 (0.503), and the covariate hazard leads 2025 (0.544). Four folds are too few for a precise variance comparison.

At a fixed top-5% budget, leading complete-year recall is 0.33–0.36. All-block activation AP averages 0.952–0.958 for leading models, showing that broad active-region identification is easier than ordering cells within active blocks. Neither number estimates field detection probability.

### Selected environmental response

The pre-2024 optimum is  $D = .28$ ,  $r = .24$ ,  $f_V = f_C = 0$ ,  $p_V = 1.0$ , and  $p_C = 1.6$ . Zero floors mean unsuitable cells are not granted artificial establishment quality; the higher

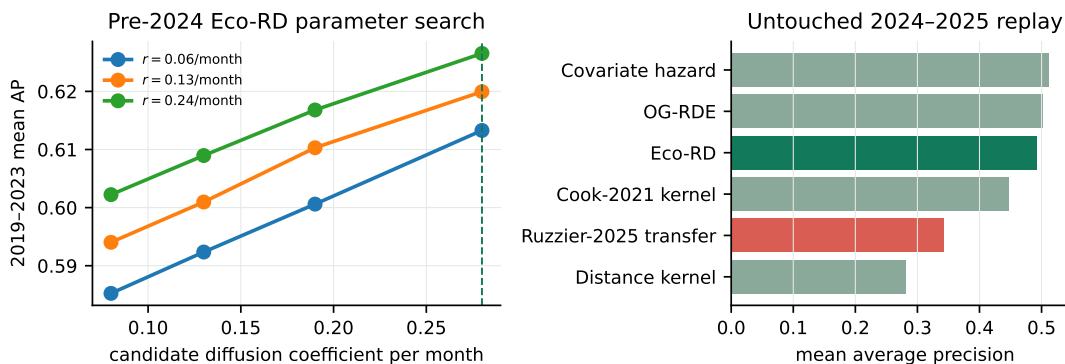
climate exponent makes middling climate more restrictive than the host surface. These are predictive parameters for this grid and state variable, not physiological estimates.

### Why the fitted model is interpretable

For one cell,  $S_i$  answers whether climate and edible vegetation jointly permit establishment. Conductivity  $k_i = .08 + .92S_i$  converts suitability into movement resistance,  $R_i = 1/k_i$ : an ideal cell has  $R = 1$ , whereas a poor cell approaches  $R = 12.5$ . The harmonic mean across a cell edge means movement is slowed by the more resistant side. Growth is separately multiplied by  $S_i$ , so a poor cell both receives pressure slowly and establishes it slowly.

### Interim 2026 slice

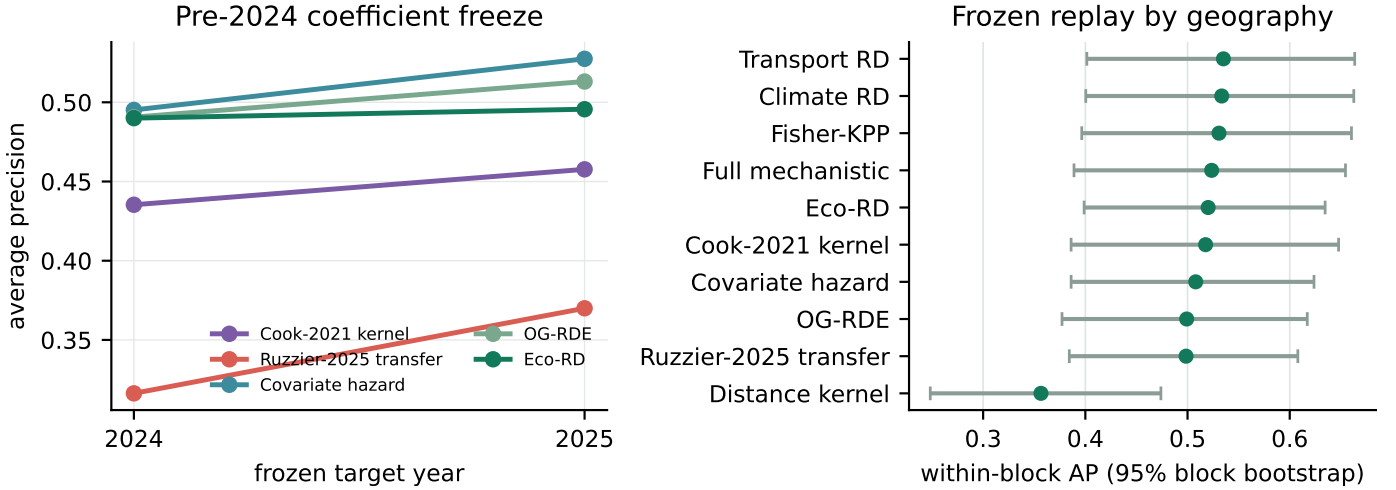
Only 24 first-report cells occur through July. Transport RD ranks first (0.202), followed by full mechanistic (0.195), the covariate hazard (0.173), Fisher–KPP (0.172), and Eco-RD (0.169). This partial, post-outcome slice is not external validation.



**Figure 4.** Left: pre-2024 parameter search, showing the best host/climate settings at each diffusion–growth pair. Right: untouched 2024–2025 mean AP. Selection and evaluation years are disjoint.



## Frozen temporal replay



**Figure 5. Left:** target-year AP after learned coefficients were fit through 2023 and frozen. Physics and literature comparators have fixed published or declared parameters. **Right:** within-block AP with 2,500 geographic-block bootstrap intervals over 31 block-years in 17 blocks. Unlike Figure 2, learned models are temporally frozen but not spatially refit.

ages 0.502. Moreover, paired geographic-block intervals for Eco-RD minus Cook (0.002, 95% interval  $-0.037$  to  $0.033$ ) and Eco-RD minus Ruzzier (0.022,  $-0.034$  to  $0.070$ ) cross zero. The annual ranking supports improvement over these transferred mechanistic comparators on this endpoint; the block analysis says available geography is insufficient to claim a uniform local advantage.

## Eco-RD improves the transferred physics baselines

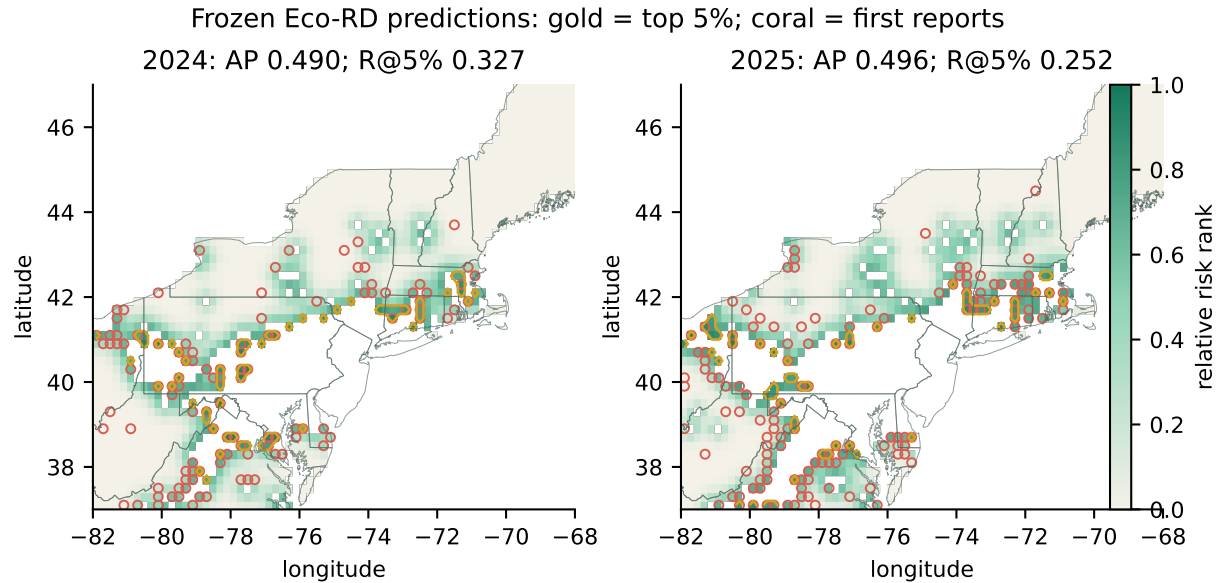
Eco-RD scores 0.490 AP in 2024 and 0.496 in 2025. Cook-2021 scores 0.435 and 0.458, so Eco-RD's mean gain is 0.046 AP, or 10.4% relative. The Ruzzier-2025 transfer scores 0.316 and 0.370; Eco-RD's mean gain is 0.150 AP, or 43.6%. Distance alone scores 0.275 and 0.288, a 75.1% relative Eco-RD improvement. Eco-RD is higher in each individual frozen year for all three comparisons, rather than relying on one favorable year.

## The claim is bounded

The covariate hazard scores 0.495 and 0.527, averaging 0.511 and remaining 3.6% above Eco-RD. OG-RDE aver-

| Model                 | 2024 AP | 2025 AP | Mean R@5% |
|-----------------------|---------|---------|-----------|
| Covariate hazard      | .495    | .527    | .285      |
| OG-RDE                | .491    | .513    | .280      |
| Eco-RD                | .490    | .496    | .290      |
| Transport RD          | .478    | .482    | .278      |
| Climate RD            | .469    | .480    | .253      |
| Fisher-KPP            | .464    | .480    | .258      |
| Full mechanistic      | .443    | .472    | .288      |
| Cook-2021 kernel      | .435    | .458    | .234      |
| Ruzzier-2025 transfer | .316    | .370    | .212      |
| Distance kernel       | .275    | .288    | .255      |

**Table 3.** Frozen replay performance. R@5% is the macro mean of annual recall in the highest-risk 5% of eligible cells.



**Figure 6.** Frozen Eco-RD maps for both complete holdout years. Gold contours enclose the fixed top-5% survey allocation; coral rings mark cells first reported in that year. The maps expose both useful concentration near the advancing front and geographically structured misses.

### What the map reveals

In 2024, the fixed 5% allocation contains 36 of 110 new-report cells; in 2025 it contains 35 of 139. Many detections fall near the modeled front, but rings beyond the gold contours show that the score misses both isolated advances and portions of the broader eastern front. A single AP value hides this distinction. The map makes false concentration and missed exploration visible to a domain expert.

### Interpolation is a hypothesis

The green field is spatially continuous even where reports are sparse because diffusion and covariate features interpolate pressure between observations. That continuity is useful for search planning, but it is modeled evidence, not an observed boundary. White masked cells are outside the eligible land domain; pale eligible cells are low-ranked, not verified absences. This separation is retained in the application: reports remain discrete, interpolated risk is labeled, and prospective growth is off by default.

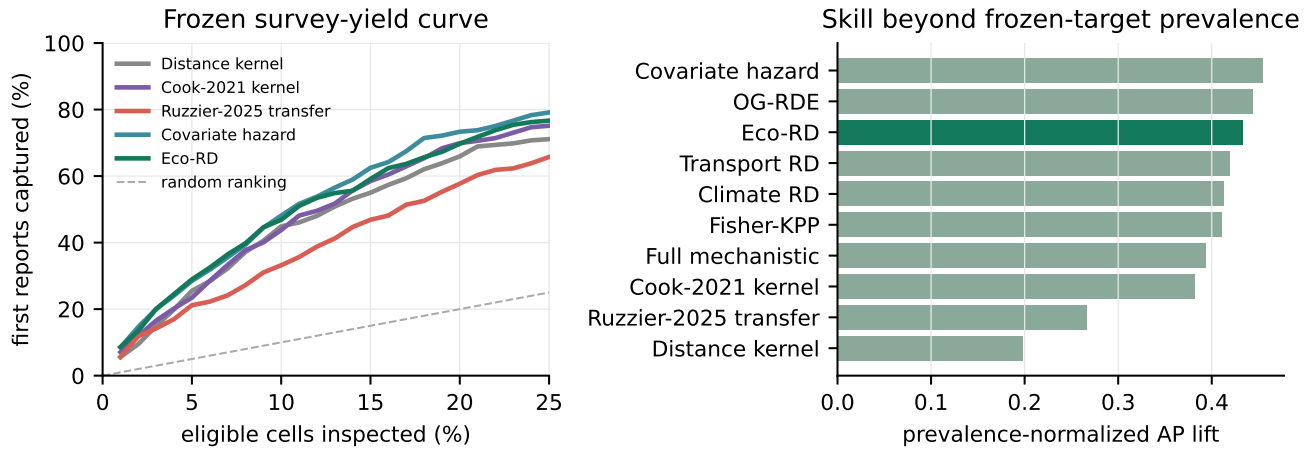
### Why the second year is harder

The 2025 top-5% recall declines even as AP rises. AP rewards ranking across the full candidate list, whereas R@5% examines one narrow operational budget. A model can improve global ordering while concentrating its very highest scores too tightly. Showing both metrics prevents a better aggregate rank from being mistaken for better performance at every decision threshold.

### What remains external

The 2024–2025 outcomes are external to coefficient fitting, but not external to the 2026 data snapshot or to analyst choices. The maps are therefore a frozen temporal stress test, not independent confirmation of occupancy. A second species or structured survey would test transfer across observation processes; a preregistered 2027 map would test real-time forecasting.

## Operational yield and model agreement



**Figure 7. Left:** mean fraction of 2024–2025 first reports captured as the survey budget expands; the diagonal is random ranking. **Right:** AP above frozen-target prevalence, normalized by the remaining attainable range.

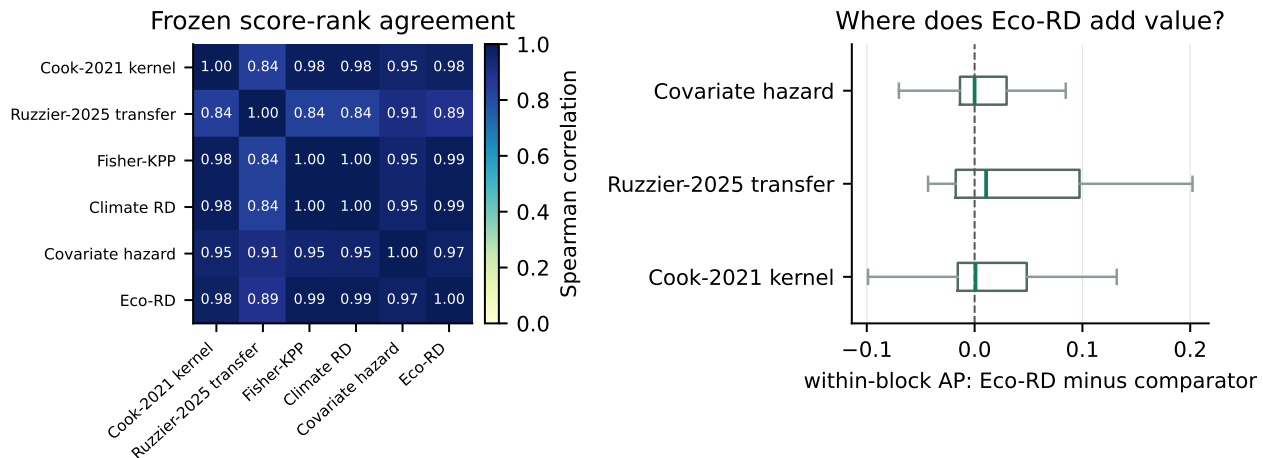
### A decision-relevant view

Average precision summarizes every threshold, but surveillance programs must choose a budget. At 5%, the frozen models capture roughly one quarter to three tenths of new reports; widening inspection increases yield smoothly rather than revealing a single defensible cutoff. This is why LanternTrace displays continuous relative risk and makes the prospective layer optional. The model can prioritize

search, but it cannot turn an arbitrary contour into an ecological boundary.

### Shared signal dominates

The frozen score fields are strongly rank-correlated (Figure 8). Their agreement reflects a common advancing-front signal carried by distance, neighborhood pressure, climate, vegetation, and the updated origin state. Eco-RD improves annual AP over both transferred published forms, but its blockwise advantages are heterogeneous. This distinction is why the paper reports both temporal rankings and geographic uncertainty.



**Figure 8. Left:** pooled Spearman rank correlation among frozen 2024–2025 score fields. **Right:** distributions of matched within-block AP differences; zero denotes no Eco-RD advantage. Agreement and heterogeneous local differences explain why an annual improvement need not occur uniformly in every region.

## Discussion

### Interpretation

The defensible result has two parts. First, cumulative evaluation is optimistic: replacing cumulative reconstruction with frontier ranking lowers raw AP by 0.368–0.470. Second, after that correction and a pre-2024 fit, climate–host physics improves the two transferred published mechanistic forms on both untouched years.

The comparison does not justify calling Eco-RD universally state of the art. Ruzzier’s raster, domain, predictors, and calibration target differ, so its score here measures a transparent transfer, not the original paper’s full performance. The study-built covariate hazard also remains higher. What is supported is narrower and useful: explicitly fitted climate and edible-host vegetation improve the transferred Cook and Ruzzier physics on this common first-report endpoint.

### Why rankings disagree

Annual AP pools cells; within-block AP conditions on blocks with both classes; all-block activation asks whether broad regions become active. A model can identify active regions yet add little local ordering skill. The block-refit analysis tests coefficient transfer while retaining origin-time state in the held block; it is not transfer to a never-monitored region.

### Research interface and evidence separation

The accompanying LanternTrace Explorer provides an interactive representation of the distinctions evaluated here. Observed first reports remain a separate evidence layer, while users may compare diffusion hypotheses, advance monthly physics steps, and inspect interpolated relative-risk contours in sparsely reported areas. Prospective playback is disabled by default, and historical assimilating animation is labeled separately from independent backtesting. These design constraints reduce the risk that modeled spread will be interpreted as observed spread.

### Limitations

**No verified absences or availability timestamps.** Pseudo-negatives contain unoccupied and undetected cells.

A 2026 snapshot is partitioned by event date, so later uploads can enter earlier reconstructed origins. Results are retrospective, not real-time historical forecasts. Independent structured validation remains essential [13].

**Static and biased covariates.** Host records are a current 21,600-record API snapshot; WorldClim is static. GBIF host occurrences reflect access and identification as well as vegetation. Survival days come from a controlled first-instar experiment, not field abundance or full life-cycle fitness.

**Coarse model and fitted predictive parameters.** The grid smooths barriers and hosts. Eco-RD’s  $D$ ,  $r$ , floors, and powers optimize first-report ranking; they are not biologically identified diffusion or demographic rates. Time-step stability is shown; grid convergence is not.

**Short temporal series.** Four exploratory folds and only two frozen targets limit temporal inference. The frozen block analysis has 31 informative block-years in 17 blocks. Block resampling covers geographic sampling variation, not covariate, observation-process, analyst-selection, or structural uncertainty.

**The freeze is not prospective.** The pre-2024 replay prevents target outcomes from changing coefficients, but it was designed after the authors had access to the 2026 snapshot. It is stronger temporal separation, not preregistration or external field validation. January–July 2026 remains post-outcome interim monitoring with 24 positives.

**No abundance or causal claim.** The state variable is relative invasion pressure. Mechanistic terms improve interpretability but do not identify true diffusion coefficients, growth rates, or transport causality from opportunistic reports.

### Future validation

The next study should freeze the present model and prospectively register 2027 predictions before reports accrue. Independent inspection routes or trap-nights would permit occupancy–detection modeling and honest probability calibration. Replacing schematic corridors with versioned road, rail, freight, and commodity-flow networks would test the human-transport mechanism directly. Leave-region-out validation against withheld states or counties would probe transfer beyond the current frontier. A hierarchical point-process or Bayesian mechanistic model could then propagate observation and process uncertainty rather than only ranking cells.

## Conclusion

Cumulative occurrence maps are appropriate summaries of reported evidence but poor targets for evaluating frontier forecasts. In this study, cumulative scoring increased raw AP by 0.368–0.470 and normalized lift by 0.384–0.511 because previously reported cells remained positive. Excluding those cells converted range reconstruction into the operationally relevant task of ranking where first reports would occur next.

In the frozen replay, Eco-RD averaged 0.493 AP. It exceeded Cook-2021 by 10.4%, the Ruzzier-2025 transfer by 43.6%, and distance alone by 75.1%, with the same direction in 2024 and 2025. However, geographic paired intervals crossed zero and the covariate hazard averaged 0.511. The supported claim is therefore improvement over transferred published mechanistic baselines on this retrospective endpoint, not universal superiority.

The principal requirement for stronger inference is prospective validation. Registering predictions before 2027 reports accrue and evaluating them against documented, independent survey effort would distinguish biological spread from discovery and permit assessment of calibrated risk rather than retrospective ranking alone.

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**Author statement.** Alex Arthur Dils developed the application, analysis, visualizations, and manuscript. Third-party data retain the source licenses documented with the public artifact. Funding, competing-interest, and permanent archive DOI statements remain to be confirmed before journal submission. All numerical claims in this manuscript are generated from the accompanying artifacts; interpretive claims are bounded by the limitations above.